Oscar Rodrigues

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Professional Summary

Data-driven Data Scientist with 4+ years of experience in analytics, data engineering, and automation, delivering business value through predictive modeling, machine learning, and data-driven insights. Proven ability to design scalable data pipelines, develop high-performing ML models, and create interactive data visualizations to optimize decision-making and drive operational improvements. Adept at leveraging tools like Python, SQL, AWS, and TensorFlow to solve complex data challenges. Collaborative team player with a track record of aligning data initiatives with business objectives to deliver impactful results.

Skills

Languages: Python, SQL, R, MATLAB
ML/AI Tools: TensorFlow, Keras, Scikit-learn, Pandas, NumPy
Data Engineering: AWS (S3, Lambda)
Visualization: Power BI, Tableau, Matplotlib
CI/CD: Git, GitHub, Docker

Professional Experience

ADAS Operations Engineer

Stellantis | March 2022 – Present

Leveraging data science and machine learning to enhance ADAS features in autonomous vehicles.

- Trained a logistic regression model to detect erratic steering behavior with 93.4% accuracy, reducing manual inspection workload by 65%
- Scraped 48 DOT websites using Python/BeautifulSoup to flag construction zones, cutting log volume by 24%
- Led the development of SOPs and internal guides, improving workflow adoption and process consistency across a 12+ person engineering team

Junior Data Analyst

SPM Automation | March 2021 – March 2022

Enabled cost-saving decisions by building data infrastructure and automating reporting systems.

- Built ETL pipelines to AWS S3 with Python and SQL, cutting reporting errors by 30% and delivery time by 50%.
- Designed Tableau dashboards for real-time production insights used by executive leadership.
- Documented business and system workflows to enhance team communication and ensure data integrity.

Education

Bachelor of Applied Science – Electrical Engineering

University of Windsor | 2020

Projects

Quantifying Vehicle Steering Discrepancies

Stellantis

- Restructured and normalized 500+ JSON log files using Pandas to enable structured analysis of steering behavior across autonomous vehicle test data.
- Visualized vehicle trajectory and signal data with Matplotlib to identify erratic steering patterns from 50+ vehicle tests, contributing to feature root-cause analysis.
- Engineered features and trained a logistic regression model to detect steering anomalies, achieving 93.4% accuracy and improving early error detection.

Proactive Road Construction Detection System

Stellantis

- Scraped construction and road closure data from 48 Department of Transportation (DOT) websites using Python (BeautifulSoup) to anticipate road hazards and reduce reliance on evasive event logs.
- Cleaned and normalized raw HTML data with Pandas, ensuring consistent, high-quality inputs for system integration and automation.
- Built an automated notification pipeline to preemptively flag high-risk construction zones, enabling preemptive disabling of ADAS features and reducing customer-reported issues.

Taylor Swift Song Data Analysis

Personal Project | `[Link to GitHub repository]`

- Extracted track data from the Spotify API using Python to analyze key audio features (tempo, energy, danceability)
- Cleaned and normalized raw data using Pandas; removed nulls and handled duplicates
- Created Matplotlib visualizations to explore trends in Taylor Swift's music over time

Traffic Sign Recognition Using Machine Learning

University Project

- Developed a convolutional neural network in Python using Keras and TensorFlow to classify traffic signs from image data
- Achieved 95.8% test accuracy across over 1,000 images, using image augmentation and dropout to improve generalization
- Demonstrated real-time object classification performance in simulation and documented findings in a project report

Certifications

Deep Learning Specialization (In Progress)

DeepLearning.AI | *Estimated August 2025*

Machine Learning Specialization `[Link to Credentials]`

Stanford University & DeepLearning.AI | July 2024

IBM Data Science Specialization `[Link to Credentials]`

IBM & Coursera | September 2023